3 Points Each

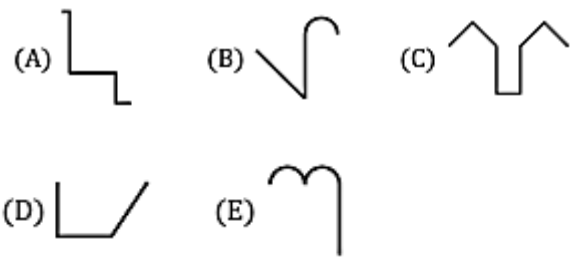
2002

1. The number 2002 read from left to right and from right to left is the same. Which number from the numbers below does not have this property?

(A) 1991
(B) 2323
(C) 2112
(D) 2222
(E) 4334

2. The picture below is a sketch of a castle. Which of the lines below does not belong to the sketch?





3. Mr. and Mrs. Kowalski have three daughters. Each of them has two brothers How many children does the Kowalski family have?

(A) 9
(B) 7
(C) 6
(D) 5
(E) 11

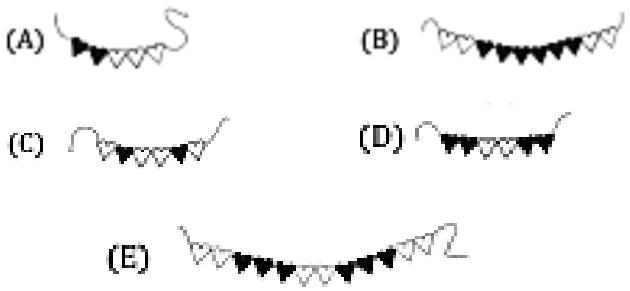
4. In which number below is the square of tens digit equal to the triple of the sum of the digits of hundreds and ones?

(A) 192
(B) 741
(C) 385
(D) 138
(E) 231

5. The product $2^2 \times 2^{2000} \times 2$ is equal to:

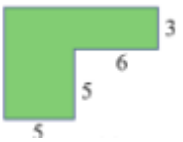
(A) 2^{400}
(B) 2^{2002}
(C 2^{2003}
(D) 2^{4002}
(E) 2^{4001}

6. On which string is the number of black hearts equal to two thirds of the number of all the hearts on the string?



7. Which of the numbers below is the greatest?

(A) $10 \times 0.001 \times 100$
(B) $0.01 \div 100$
(C) $100 \div 0.01$
(D) $10{,}000 \times 100 \div 10$
(E) $0.1 \times 0.01 \times 10{,}000$

8. 

The area of the figure in the picture is equal to:

(A) 43
(B) 88
(C) 58
(D) 30
(E) 15

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9. The area of a certain rectangle is equal to 1 m². What is the area of a triangle that was cut off from the rectangle along the line connecting the midpoints of the two adjacent sides?

(1m10 dm = 100 cm)

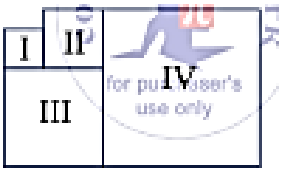
(A) 33 dm²
(B) 25 dm²
(C) 40 dm²
(D) 3,750 cm²
(E) 1,250 cm²

10. We subtracted the smallest three-digit number with all different digits from the greatest three-digit number with all different digits. The result was:

(A) 864
(B) 885
(C) 800
(D) 899
(E) a different number

4 Points Each

11. Figures I, II, III, and IV are squares. The perimeter of square I is equal to 16 m, and the perimeter of square II is equal to 24 m. The perimeter of square IV is equal to:

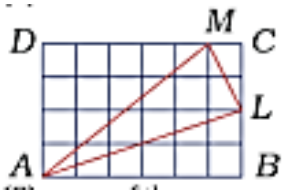


(A) 56 m
(B) 60 m
(C) 64 m
(D) 72 m
(E) 80 m

12. One medal can be cut out from a gold square plate. If four medals are made from four plates, the remaining pieces of those four plates can be used to make one more plate. What is the largest number of medals that can be formed when 64 plates are used?

(A) 85
(B) 64
(C) 80
(D) 84
(E) 100

13. Rectangle ABCD (see the picture) is made of 24 little squares with the length of each side equal to 1. What is the area of the triangle ALM?



(A) 5
(B) 6
(C) 7
(D) 8
(E) none of these

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14. In the picture below the coordinates of points A and Bare indicated. What are the coordinates of points Cand D if |AB|=2|BC| and |BC| = 2 |CD|?

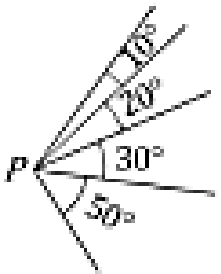


(A) 24 and 32
(B) 24 and 28
(C) 24 and 26
(D) 22 and 24
(E) 22 and 23

15. Mark has 9 sticks with the lengths of 1 dm, 2 dm, 3 dm, 4 dm, 5 dm, 6 dm, 7 dm, 8 dm, and 9 dm. Using the sticks, he builds triangles with each side made with one stick. How many triangles with a side of 1 dm can be built using these sticks?

(A) 6
(B) 3
(C) 2
(D) 1
(E) 0

16. How many convex angles with different measures are made by the rays with P as the starting point (see picture)? (A convex angle is an angle that is larger than a zero angle and smaller than a straight angle.)



(A) 4
(B) 6
(C) 8
(D) 10
(E) 11

17. How many different three-digit numbers divisible by 25 can be made with the digits 0, 3, 5, and 7 if the digits can be repeated?

(A) 16
(B) 9
(C) 81
(D) 64
(E) 3

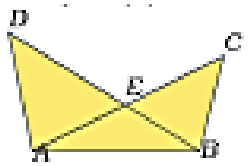
18. Each of the boys, Mike, Nate, Oliver, and Paul, has exactly one of the following animals: a cat, a dog, a goldfish, and a canary. Nate has a pet with fur. Oliver has a pet with four legs, Paul has a bird, and Mike and Nate don't like cats. Which of the following sentences is not true?

(A) Oliver has a dog.
(D) Oliver has a cat.
(B) Paul has a canary.
(C) Mike has a goldfish.
(E) Nate has a dog.

19. The day after his birthday Johnny said: "The day after tomorrow will be Thursday." On what day of the week did Johnny have his birthday?

(A) on Monday
(B) on Tuesday
(C) on Wednesday
(D) on Thursday
(E) on Friday

20. The area of triangle ABD is equal to 12, the area of triangle ABC is equal to 15 and the area of triangle ABE is equal to 4 (see the picture). What is the area of pentagon ABCED?



(A) 19
(B) 31
(C) 23
(D) 27
(E) 35

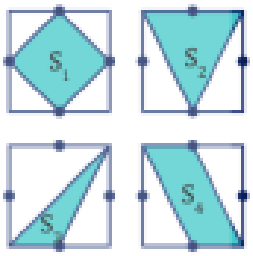
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5 Points Each

21. The weight of each possible pair of boys from a group of 5 boys was recorded. The following results were obtained: 90 kg, 92 kg. 93 kg, 94 kg, 95 kg, 96 kg, 97 kg, 98 kg, 100 kg, and 101 kg. The total weight of all five boys equals:

(A) 225 kg
(B) 230 kg
(C) 239 kg
(D) 240 kg
(E) 250 kg

22. There are four congruent squares. In each of them the midpoints of the sides are indicated and regions with areas S1, S2, S3, and S+ are shaded (see the picture). Which statement below is true?



(A) $S_3 < S_4 < S_1 = S_2$
(B) $S_3 < S_1 = S_2 = S_4$
(C) $S_3 < S_1 = S_4 < S_2$
(D) $S_3 < S_4 < S_1 < S_2$
(E) $S_4 < S_3 < S_1 < S_2$

23. You count from 1 to 100 and clap when you say the multiples of the number 3 and when you say numbers that are not multiples of 3 but have 3 as the last digit. How many times will you clap your hands?

(A) 30
(B) 33
(C) 36
(D) 39
(E) 43

24. A cyclist went up the hill with a speed of 12 km/h and went down the hill with a speed of 20 km/h. The ride up the hill took him 16 minutes longer than the ride down the hill. How many minutes did it take the cyclist to go down the hill?

(A) 24
(B) 40
(C) 32
(D) 16
(E) 28

25. The letters P, Q, R, and S indicate the total weight of the figures drawn next to them as shown below.



It is known that any two figures of the same shape have the same weight. If P <Q<R, then:

(A) P < S < Q
(B) Q < S < R
(C) S < P
(D) R < S
(E) R – S

26. Ada has 14 gray marbles, 8 white marbles, and 6 black marbles in a bag. What is the least number of marbles she has to take out of her bag with her eyes closed to be sure that she took at least one marble of each color?

(A) 23
(B) 22
(C) 21
(D) 15
(E) 9

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27. A certain computer virus destroys computer memory. On the first day it destroyed of the memory of a certain computer. On the second day it destroyed of the memory remaining after the first day. On the third day it destroyed of the memory remaining after the second day, and on the fourth day it destroyed of the memory remaining after the third day. What 5 part of all the computer's memory was left after those four days?

(A) $\frac{1}{5}$

(B) $\frac{1}{6}$

(C) $\frac{1}{10}$

(D) $\frac{1}{12}$

(E) $\frac{1}{24}$

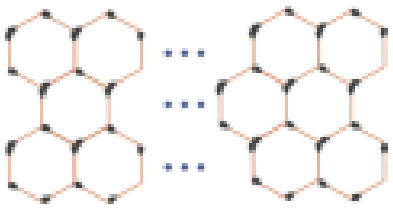
28. What is the greatest value of the sum of the digits of the number made from the sum of the digits of a three-digit number?

(A) 9
(B) 10
(C) 11
(D) 12
(E) 18

29. 32 players were competing in a chess competition. The competition was taking place in stages. In each stage all the players were divided into groups of four. In each of these groups every player played once with each of the other players. Two players from the group went to the next level and the other two players were out of the competition. After the stage in which the four last players played, the two top players played an additional game. How many games were played during the whole competition?

(A) 49
(B) 89
(C) 91
(D) 97
(E) 181

30. A net with 32 hexagonal spaces in three rows was made out of matches: as shown below.



How many matches were used to make this net?

(A) 123
(B) 124
(C) 125
(D) 120
(E) 121

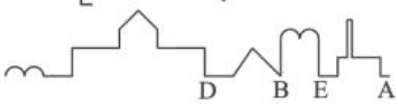
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Answers key:

1. B
2. C
3. D
4. E
5. C
6. D
7. D
8. C
9. E
10. B
11. C
12. A
13. C
14. E
15. E
16. C
17. B
18. A
19. A
20. C
21. C
22. B
23. D
24. A
25. A
26. A
27. A
28. B
29. C
30. A

for purchaser's use only

1. (B) 2323
The first and the last digits are different.

2. (C) 
Below four lines are identified as segments of the sketch.
(A)  (B)  (C)  (D)  (E) 


Line (C) does not belong to the sketch.

3. (D) 5
The two boys are brothers to all three daughters. $3 + 2 = 5$, so Mr. and Mrs. Kowalski have 5 children, 3 girls and 2 boys.

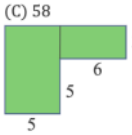
4. (E) 231
$3^2 = 3 \times (2 + 1)$
For other numbers: $9^2 > 3 \times (1 + 2)$, $4^2 < 3 \times (7 + 1)$, $8^2 > 3 \times (3 + 5)$, and $3^2 < 3 \times (1 + 8)$.

5. (C) 2^{2003}
We add the exponents when multiplying powers of the same base, so $2^2 \times 2^{2000} \times 2 =$
$= 2^2 \times 2^{2000} \times 2^1 = 2^{2+2000+1} = 2^{2003}$.

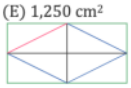
6. (D) 
There are 4 black hearts on this string of 6 hearts and $4 = \frac{2}{3} \times 6$. For the other strings the equation $b = \frac{2}{3} \times a$ (b = the number of black hearts, a = the number of all hearts) is false.
(A)  $b = 2, a = 5$
(B)  $b = 6, a = 10$
(C)  $b = 2, a = 6$
(E)  $b = 6, a = 12$

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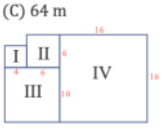
7. (D) $10{,}000 \times 100 \div 10$
When multiplying by 10 move the decimal point to the right, and when multiplying by a decimal move the decimal point to the left. When dividing, move the decimal point the other direction.
(A) $10 \times 0.001 \times 100 = 1000 \times 0.001 = 1$
(B) $0.01 \div 100 = .0001$
(C) $100 \div 0.01 = 10{,}000$
(D) $10{,}000 \times 100 \div 10 = 100{,}000$
(E) $0.1 \times 0.01 \times 10{,}000 = 10$
Of these, the greatest number is 100,000, answer (D).

8. (C) 58


The black vertical line divides the figure into two rectangles. The left rectangle has the dimensions $(5 + 3) \times 5$ and the right rectangle has the dimensions 3×6, so the combined area is $(8 \times 5) + (3 \times 6) = 58$.

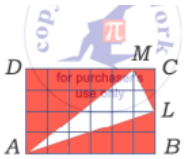
9. (E) 1,250 cm²

A rectangle with all six pairs of midpoints of its sides connected is shown. The rectangle is divided into 8 equal triangles, so the area of each triangle is $\frac{1}{8} \times 1$ m². 1 m = 100 cm, so 1 m² = 100 cm × 100 cm = 10,000 cm² and $\frac{1}{8} \times 10{,}000$ cm² = 1,250 cm², so each triangle (including the cut-off one) has an area of 1,250 cm².

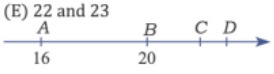
10. (B) 885
The smallest three-digit number with all different digits is 102 and the greatest three-digit number with all different digits is 987. The difference between them is 885.

11. (C) 64 m


Each side of the square I has the length of 16 m ÷ 4 = 4 m. Each side of the square II has the length of 24 m ÷ 4 = 6 m. Hence, the top side of the square III, and any other side of this square, has a length of (4 m + 6 m) = 10 m. The left side of the square IV has a length of (6 m + 10 m) = 16 m. Therefore, the perimeter of the square IV is 4 × 16 m, which is 64 m.

12. (A) 85
64 = 16 × 4, so there are 16 groups of 4 plates each. From each group we can form 4 medals and 1 new plate, so we can form 16 × 4 medals = 64 medals and 16 new plates. Put the 64 medals aside. 16 = 4 × 4, so there are 4 groups of 4 new plates each. From each group we can form 4 medals and one extra plate, so we can form 4 × 4 medals = 16 medals and 4 extra plates. From the last 4 plates we can form 4 medals and 1 plate. From the last plate 1 medal can be cut out, so the largest number of medals that can be formed from 64 plates is 64 + 16 + 4 + 1 = 85.

13. (C) 7
The area of rectangle ABCD is 24. If we remove the three right triangles, we will get the area of triangle ALM. The area of any right triangle is half of the product of the lengths of its legs, so the area of triangle ABL is $(6 \times 2) \div 2 = 6$, the area of triangle LCM is $(2 \times 1) \div 2 = 1$, and the area of triangle MDA is $(5 \times 4) \div 2 = 10$. 6 + 1 + 10 = 17 is the total area removed from the rectangle, so the area of triangle ALM is 24 − 17 = 7.


14. (E) 22 and 23


$|AB| = (20 - 16) = 4$, $|BC| = 4 \div 2 = 2$ and $|CD| = 2 \div 2 = 1$. C is to the right of B, so the coordinate of C is 20 + 2 = 22. D is to the right of C, so the coordinate of D is 22 + 1 = 23.

15. (E) 0
In any triangle, the sum of the lengths of any two sides is strictly greater than the length of the third side. If m = the length of the medium side and ℓ = the length of the longest side, then $m < \ell$. If both m and ℓ are natural numbers, then $(m + 1) \le \ell$ since m + 1 is the smallest natural number greater than m. If 1 is the shortest side of a triangle (with the other two sides being m and ℓ), then $m + 1 > \ell$ but $(m + 1) \le \ell$, so the stick of the length 1 cannot be a side of any triangle built from three of the given sticks.

16. (C) 8
A convex angle is an angle greater than 0° but smaller than 180°.
50° + 30° + 20° + 10° = 110° < 180°, so we can pick any two rays to have a convex angle. There are 5 rays and we can pick two at a time in 10 different ways. The measures of the 10 angles are: 50°, 50° + 30° = 80°, 50° + 30° + 20°= 100°, 50° + 30° + 20° + 10° = 110°; 30°, 30° + 20°= 50°, 30° + 20° + 10° = 60°; 20°, 20° + 10° = 30°; and 10°. Among them there are 8 different measures.

17. (B) 9
A number is a multiple of 25 if it ends with 00, 25, 50, or 75. Since 2 is not one of the given digits, we are left with three possibilities: 00, 50, or 75. The first digit can only be 3, 5, or 7, so there are 3 × 3 numbers that are 3-digit numbers divisible by 25 and use only digits 0, 3, 5, 7. Here is a complete list of such numbers: 300, 350, 375; 500, 550, 575; 700, 750, 775.

18. (A) Oliver has a dog.
Nate, who has a pet with fur but doesn't like cats, has to have a dog. Because Nate owns a dog, the sentence, "Oliver has a dog," is not true. Actually, Oliver owns a cat. The other sentences are true.

19. (A) on Monday
If the day after tomorrow is Thursday, then tomorrow is Wednesday, and today is Tuesday. It is one day after Johnny's birthday, so he had his birthday on Monday.

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20. (C) 23
The area of triangle BCE is the area of triangle ABC minus the area of triangle ABE. $15 - 4$ is 11, so the area of triangle BCE is 11. Similarly, the area of triangle EDA is $12 - 4 = 8$. The area of the pentagon is $4 + 11 + 8 = 23$.

21. (C) 239 kg
Each boy was weighed with each of the other 4 boys, so each boy was weighed 4 times.
Add the ten results and divide by 4 to see the total weight of all five boys.
$90 + 92 + 93 + 94 + 95 + 96 + 97 + 98 + 100 + 101 =$
$(90 + 100) + (92 + 98) + (93 + 97) + (94 + 96) + (95 + 101) = 4 \times 190 + 196 = 956.$
$956 \div 4 = 239$, so the total weight of all five boys equals 239 kg.

22. (B) $S_3 < S_1 = S_2 = S_4$
Let S be the area of each square.



The whole square is divided into 8 equal triangles and 4 of them are shaded, so $S_1 = ½S$.



The whole square is divided into 4 equal triangles and 2 of them are shaded, so $S_2 = ½S$.



The diagonal is dividing the square into 2 equal parts. One part consists of the shaded triangle and the right triangle at the lower right corner. Both triangles have the same height and their bases have the same length (half of the square side), so they have the same area which is ¼S. Hence, $S_3 = ¼S$.



The whole square is divided into 4 equal triangles and 2 of them are shaded, so $S_4 = ½S$.

S_3 has an area that is smaller and all the other areas are the same, so $S_3 < S_1 = S_2 = S_4$ is the only true statement.

23. (D) 39
The multiples of 3 between 1 and 100 are: $1 \times 3, 2 \times 3, 3 \times 3, 4 \times 3, \ldots, 33 \times 3$.
There are 33 of them. Numbers which are not multiples of 3 but have the last digit 3 are: 13, 23, 43, 53, 73, and 83. There are 6 of them. Together, there are $33 + 6 = 39$ of them, so you will clap your hands 39 times.

---

24. (A) 24
The cyclist went up and down the same hill, so the distance travelled up and down is the same. The formula for calculating distance travelled is speed multiplied by time.
The time it took the cyclist to travel down the hill was t minutes, and the time it took to go up was $t + 16$ minutes. Using the formula for distance travelled, the distance of going uphill is 12km/h times $t + 16$ minutes, and the distance going downhill is 20km/h $\times$ t minutes. Since the distance of uphill and downhill are the same, $12 \times (t + 16) = 20\,t$
Solving, we get

$$12 \times t + 12 \times 16 = 20t$$
$$12t + 192 = 20t$$
$$192 = 8t$$
$$t = 24$$

This mean that it takes the cyclist 24 minutes to travel down the hill.

25. (A) P < S < Q

The weight of  is P and the weight of  is Q. Each contains one square and P < Q, so we can remove each square to see that weigh less than . Hence,  weighs less than  , or .

Add  to both sides of the last inequality. Then  , so P < S.
Use again. Add  to both sides to get . Hence, S < Q.
In conclusion, P < S < Q. The other statements are false since each of them contradicts the true statement that P < S < Q < R.

26. (A) 23
With 14 gray marbles, 8 white marbles, and 6 black marbles it may happen that among the first 22 marbles ($22 = 14 + 8$) taken out there are no black marbles. To make sure that among randomly selected marbles at least one is black, Ada must take out at least 23 marbles. Similarly, to have at least one white marble among randomly selected marbles Ada must take out at least 21 marbles ($21 = 14 + 6 + 1$). To have at least one gray marble among randomly selected marbles Ada must take out at least 15 marbles ($15 = 8 + 6 + 1$). Hence, to have at least one marble of each color Ada must take out at least 23 marbles.

27. (A) $\frac{1}{5}$
On the first day, the virus destroyed $\frac{1}{2}$ of the memory, so there was $\frac{1}{2}$ of the original memory left at that point. On the second day, it destroyed $\frac{1}{2} \times \frac{1}{3} = \frac{1}{6}$ of the original memory, so there was $\frac{1}{2} - \frac{1}{6} = \frac{3}{6} - \frac{1}{6} = \frac{2}{6} = \frac{1}{3}$ of the original memory left. On the third day, the virus destroyed $\frac{1}{3} \times \frac{1}{4} = \frac{1}{12}$ of the original memory, so there was $\frac{1}{3} - \frac{1}{12} = \frac{4}{12} - \frac{1}{12} = \frac{3}{12} = \frac{1}{4}$ of the original memory left. On the fourth day, the virus destroyed $\frac{1}{4} \times \frac{1}{5} = \frac{1}{20}$ of the original memory, so there was $\frac{1}{4} - \frac{1}{20} = \frac{5}{20} - \frac{1}{20} = \frac{4}{20} = \frac{1}{5}$ of all the computer's memory left after those four days.

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28. (B) 10

If xyz is a given 3-digit number, then $1 \le x \le 9$ and $0 \le y, z \le 9$. The sum $s = x + y + z$ is a number between $1 + 0 + 0 = 1$ and $9 + 9 + 9 = 27$. We add the digits of s searching for the largest result. If s is a one-digit number, then the sum of its digits is s. The largest sum among them is 9. Next we have $s = 10, 11, 12, \ldots, 19$, where the largest sum is $1 + 9 = 10$. In the last group of 20, 21, 22, ..., 27, the largest sum is $2 + 7 = 9$.
10 is the greatest value for the sum of the digits of s. It happens any time $s = x + y + z = 19$. There are 45 such numbers. We may list them in a systematic way.
199; 289, 298; 379, 388, 397; 469, 478, 487, 496; 559, 568, 577, 586, 595; 649, 658, 667, 676, 685, 694; 739, 748, 757, 766, 775, 784, 793; 829, 838, 847, 856, 865, 874, 883, 892; 919, 928, 937, 946, 955, 964, 973, 982, 991.

29. (C) 91

In the first stage there were 8 groups each with 4 players. Since each of the players played a game with each of three other players, each group played 6 games, so in the 8 groups the players played $8 \times 6 = 48$ games.
48 is the number of games in the first stage.
The second stage started with 16 players (2 from each of the initial 8 groups) so there were 4 groups with 4 players each. Again, each group of 4 players played 6 games, so in the 4 groups the players played $4 \times 6 = 24$ games.
24 is the number of games in the second stage.
The third stage started with 8 players (2 from each of the previous 4 groups) so there were 2 groups with 4 players each. Each group played 6 games, so in the 2 groups the players played $2 \times 6 = 12$ games.
12 is the number of games in the third stage.
The fourth stage started with 4 players (2 from each of the last 2 groups).
They played 6 games and the two top players played 1 more game, so during the whole competition $48 + 24 + 12 + 6 + 1 = (48 + 12) + (24 + 6) + 1 = 91$ games were played.

30. (A) 123

The first row and the third one have the same number of hexagons but the middle row has one hexagon less. Together there are 32 hexagons, so the first and the third row have 11 hexagons each and the middle row has 10 hexagons. It is easy to count all non-vertical matches in the first and the third row since these matches do not overlap. There are $11 \times 4 + 11 \times 4 = 88$ non-vertical matches in all three rows since the non-vertical matches of the middle row have been already counted.
The first row has 11 hexagons, so it has 12 vertical matches. The same is true for the third row. The second row has 10 hexagons, so it has 11 vertical matches.
Together, $88 + 12 + 12 + 11 = 123$, so 123 matches were used to make this net.

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